
Closing Evidence-Blindness in Radiology Report Generators via Dual-Adversarial Causal-Faithfulness Training

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Abstract

Vision-language models that draft radiology reports often produce text that does not reflect the underlying chest X-ray. Diagnostic studies have documented this evidence-blindness, and existing fixes wrap the trained model in inference-time filters or conformal selection without changing what the model has learned. We intervene at training time. We present dual-adversarial causal-faithfulness training, a LoRA fine-tune of MedGemma-4B-IT with two ingredients: a symmetric adversarial filter that downweights training rows that either a text-only or an image-only predictor can already solve, and a causal-faithfulness loss that penalises the model whenever the probability of the gold token fails to drop under image masking. We pair the recipe with IMG_correct, a refinement of the standard image-masking gap that excludes a known failure mode in which a consistency-loss-trained model flips its top prediction under masking without losing support for the right answer. On public-data OpenI chest radiographs, the method drives IMG_correct from 2.38 ± 0.26 to 74.96 ± 0.83 pp at matched token-level accuracy across three seeds. A cross-counterfactual evaluation reports honestly that the gain is largely specific to the trained counterfactual: cropping and centre occlusion collapse it. A retroactive IMG_correct audit of 18 prior verifier cells finds induced-inversion contributions uniformly small, so the methodological refinement verifies prior numbers rather than overturning them.

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1. Introduction

Multimodal medical AI systems produce text that disregards the visual input: Restrepo et al. (2025) document this across six open-source chest-X-ray vision-language models with Selective Modality Shifting. The phenomenon mirrors the partial-input shortcuts of NLI (Gururangan et al., 2018; Poliak et al., 2018) and the multimodal-reward-model shortcuts of Li et al. (2025a). Two gaps motivate this work. First, existing mitigations are either measurement-only (the diagnostic literature) or frozen-predictor wrappers (conformal selection and post-hoc filtering; Gui et al., 2024; Li et al., 2025b; Mohri & Hashimoto, 2024); none modifies what the underlying generator has learned. Second, the only training-time mitigation at scale — the Shortcut-Failure Coefficient (SFC) of Li et al. (2025a) — operates only on the text-only side. We close both gaps.

Contributions. (C1) Dual-adversarial causal-faithfulness training: a LoRA fine-tune of MedGemma-4B-IT (Sellersgren et al., 2025; Hu et al., 2022) that combines a symmetric text + image adversarial filter (the image-only branch is the incremental axis relative to Li et al., 2025a) with a token-wise correctness-gain margin loss on gold tokens. (C2) IMG_correct: a methodological refinement of the image-masking gap (IMG) that excludes the induced inversion failure mode in which a consistency-loss-trained model flips its top prediction under masking without losing support for the right answer. We apply IMG_correct retroactively to prior verifier checkpoints and find induced-inversion contributions are uniformly small (Section 4). We additionally run a cross-counterfactual evaluation that honestly reports the gain is largely specific to the trained counterfactual.

2. Methods

We fine-tune MedGemma-4B-IT (Sellersgren et al., 2025) with LoRA (Hu et al., 2022) rank 16 on q/k/v/o projections (11.9 M trainable parameters; full hyperpa-

rameters in Appendix B.4). The training objective is

$$\mathcal{L}(x) = w(x) \cdot \mathcal{L}_{\text{SFT}}(x) + \lambda_{\text{faith}} \cdot \mathcal{L}_{\text{faith}}(x), \quad (1)$$

with $\lambda_{\text{faith}} = 0.3$.

Dual adversarial filter. The per-row weight is $w(x) = \min(w_{\text{text}}(x), w_{\text{image}}(x))$, combining two shortcut detectors trained independently: a RoBERTa-large text-only cross-encoder (reusing prior verifier infrastructure [citation withheld for double-blind review]) and a BiomedCLIP (Zhang et al., 2024) image-only MLP head. Rows on which either branch’s gold-class probability exceeds 0.7 are downweighted to $w_{\text{min}} = 0.2$. On the 20,560-row training pool, the image-only branch catches 6.5% of rows the text-only branch misses — the incremental axis relative to Li et al. (2025a) (decomposition in Appendix B.2).

Causal-faithfulness loss. The naive consistency loss $\text{ReLU}(m - \text{KL}(p_{\text{full}} \| p_{\text{alt}}))$ admits a degenerate optimum: the model can satisfy the margin by being arbitrarily wrong under masking. We replace symmetric KL with a token-wise correctness-gain margin on gold tokens,

$$\mathcal{L}_{\text{faith}}(x) = \frac{1}{|T|} \sum_{t \in T} \text{ReLU}(m - (\log p_{\text{full}}(y_t) - \log p_{\text{alt}}(y_t))), \quad (2)$$

which couples the divergence requirement to gold-token correctness on the full input ($m = 0.3$ nats). Image masking uses the per-channel mean of the batch rather than a literal zero tensor: post-normalisation zeros are out-of-distribution adversarial input, whereas the per-channel mean is the model’s neutral grey image (Appendix B.1).

IMG_correct. The standard image-masking gap counts every $\arg \max$ flip under masking as image-grounded behaviour. A consistency-loss-trained model can drive image-zeroed accuracy to zero by always flipping its $\arg \max$ even when its gold-class probability barely drops. IMG_correct counts only examples where (a) the full input is correct, (b) the masked input is incorrect, and (c) the gold-class probability drops by at least margin τ . We report induced inversion as the per-example contribution caught by IMG but excluded by IMG_correct (formal derivation in Appendix B.3).

3. Experimental Setup

Data. Training and evaluation use public-data sources only: OpenI (Demner-Fushman et al., 2016) (10,839 rows after text-target filtering; OpenI test split with 1,587 examples for in-distribution eval) and

Table 1. Headline IMG_correct on the OpenI test split (token-level, $n = 33,173$). Seed-mean $\pm$ standard deviation over three seeds. Induced-inversion (Inv) contributions are essentially zero on all faith-loss configurations.

Config	IMG_corr	IMG	Inv	Acc (%)
sft_only	2.38 ± 0.26	0.36	0.40	74.31
sft_dual	2.10 ± 0.24	0.89	0.58	75.41
sft_faith	74.11 ± 0.26	74.04	0.01	75.61
sft_full	74.96 ± 0.83	74.96	0.01	75.48
sft_full_sfc	75.42 ± 0.28	75.43	0.01	75.44

ChestX-Det10 silver targets (Liu et al., 2020) for cross-site eval.

Configurations. Five training configurations $\times$ three seeds: sft_only (SFT baseline), sft_dual (filter only), sft_faith (faith loss only), sft_full (both; headline configuration), and sft_full_sfc (replaces the discrete dual filter with continuous SFC weights from Li et al., 2025a).

Metrics. IMG_correct at $\tau = 0.1$; standard IMG; induced inversion; token-level accuracy on gold target tokens.

4. Results

Headline (Table 1). The pre-registered ≥ 5 pp improvement is exceeded by an order of magnitude on three seeds $\times$ three faith-loss configurations. The full method sft_full lifts IMG_correct by +72.58 pp at +1.17 pp accuracy over the SFT-only baseline. The dual-filter-only configuration is statistically indistinguishable from baseline; the faith loss is the load-bearing component (sft_faith alone gives +71.73 pp). Induced inversion is essentially zero across all faith-loss configurations.

Cross-counterfactual eval (honesty check). We evaluate sft_full (seed 42) under three counterfactuals the faith loss did not train on. Under noise replacement IMG_correct stays at 62.20 pp. Under crop (9% true-image patch retained) it collapses to 4.81 pp. Under centre occlusion (25% of the image area zeroed) it collapses to 1.11 pp. The trained-mean reference is 74.00 pp on this seed. The gap-to-baseline on crop is $4.81 - 2.80 = 2.01$ pp — an order of magnitude smaller than the headline gap on the trained counterfactual (cf. Appendix D). The model tightly fits the trained per-channel-mean distribution but does not transfer to localised counterfactuals.

Cross-site transfer. On 1,122 ChestX-Det10 silver-target rows, `sft_full` attains `IMG_correct` = 54.62 pp at 54.65% accuracy; the matched SFT-only baseline attains 5.12 pp. The cross-site gap of 49.5 pp is $\approx 68\%$ of the in-distribution gap, consistent with partial transfer (supp. appendix).

`IMG_correct` retroactive audit. We apply `IMG_correct` to 18 previously reported verifier configurations (nine v5/v6 headline cells + nine v5 ablations). Induced-inversion contributions are uniformly ≤ 1.31 pp absolute ($\leq 13.2\%$ of the corresponding IMG). The pre-registered $\geq 30\%$ hypothesis is refuted on every cell measured. The `abl_no_consist` cell — a v5 variant with the consistency loss removed — collapses `IMG_correct` to 3.60 pp while also reducing induced inversion, the opposite of what the pre-registered concern would predict (full table in Appendix C).

5. Discussion

What `IMG_correct` shows about prior systems. Prior IMG numbers on the v5/v6 verifier family were largely measuring real image-grounded behaviour. The `IMG_correct` contribution is therefore (a) a verification that increases confidence in prior IMG claims under a strict per-example correctness criterion, and (b) a default-to-report-alongside-IMG diagnostic that detects the failure mode if less calibrated training recipes were to introduce it.

Cross-counterfactual partial generalisation. The +72.58 pp headline measures responsiveness to the trained counterfactual, with strong transfer to similar untrained globally-uniform counterfactuals (noise) and weak transfer to localised counterfactuals (crop, occlude). The honest read: training-time counterfactual interventions can tightly fit the in-distribution counterfactual; a more diverse training mix is the natural follow-up.

Public-data only. Reproducibility in medical imaging is limited by access barriers to MIMIC-CXR-credentialed data (Johnson et al., 2019). Our public-data commitment (OpenI training + ChestX-Det10 silver cross-site) means any researcher with comparable cloud compute can re-run the pipeline. This costs scale (MIMIC-CXR has $\sim 50\times$ more rows), but the methodology generalises.

6. Limitations and Conclusion

Limitations. (L1) The faith loss and `IMG_correct` measure overlapping per-token quantities; the cross-counterfactual eval quantifies this caveat empirically. (L2) OpenI-only training; cross-site only on silver targets. (L3) Filter calibration was trained for our prior verifier task and reused; the conceptually-aligned per-task filter is an approximation. (L4) Single backbone; the retroactive audit is on our own verifier checkpoints only.

We presented dual-adversarial causal-faithfulness training and the `IMG_correct` methodological refinement. Pre-registered empirical claims, tested transparently: (i) ≥ 5 pp `IMG_correct` improvement: hit by an order of magnitude. (ii) $\geq 30\%$ of v5.3’s IMG as induced inversion: refuted (1.1%). (iii) the dual filter dominates the faith loss: inverted (faith loss is load-bearing). `IMG_correct` lands independently of the empirical results and is a reusable audit tool for any IMG-style metric. Counterfactual evidence-sensitivity testing should precede aggregate-accuracy reporting for any multimodal medical generator.

Table 2. Four-cell decomposition of which filter branch flags each training row. On the 20,560-row training pool, the image-only branch catches an additional 6.5% of rows that the text-only branch alone would miss; this is the symmetric-extension contribution axis relative to Li et al. (2025a).

Filter outcome	Count	% of pool
Both filters caught (unimodally solvable on either axis)	15,375	74.8%
Text-only caught, image-only passed	2,156	10.5%
Image-only caught, text-only passed (novelty axis)	1,329	6.5%
Neither caught (genuinely multimodal)	1,700	8.3%

A. Reproducibility Checklist

Met: (i) Training data sources are public (OpenI; ChestX-Det10 silver-target for cross-site eval; PadChest-GR is named as deferred cross-site target; MIMIC-CXR is not used). (ii) Hyperparameters in Appendix B.4; per-configuration YAML in the released codebase. (iii) 30 unit tests pass on CPU. (iv) Modal H100 80 GB infrastructure documented; all entrypoints documented. (v) Per-fine-tune training-step counts, save-every-steps schedule, and resume-from-checkpoint logic in the released code.

Deferred to extended version: code-release URL, weights-release URL (anonymized for double-blind review; filled at camera-ready); cross-site eval on PadChest-GR (no silver targets available in present infrastructure); three-seed averaging on noise/crop/occlude cross- counterfactuals (single-seed in the main paper’s Results section, three-seed values in App. D); mech-interp sweep validating a gradient- norm monitor’s AUROC.

B. Extended Methods

B.1. Faith loss derivation and implementation

The naive consistency loss $\mathcal{L}_{\text{KL}} = \text{ReLU}(m - \text{KL}(p_{\text{full}} \| p_{\text{alt}}))$ admits a degenerate optimum: a model that always emits a uniform distribution under masking satisfies any finite KL margin without using the image. The token-wise correctness-gain margin replaces the distributional divergence with a per-token signal coupled to gold-token correctness on the full input. The implementation runs three forward passes per training step (full, image-zeroed, image-flipped on laterality rows) and exposes a flipped-row mask that zeroes the flipped contribution on non-laterality rows. Image-zeroing uses the per-channel mean of the batch’s pixel values, not a literal zero tensor: a literal zero is, after the processor’s normalisation, a specific out-of- distribution input the model has never seen at training time. The per-channel mean produces a uniform “neutral” image at the model’s own normalisation, which is what we want when we say “the image is absent.”

B.2. Dual filter four-cell decomposition

After the subsequent text-target filter restricts training to OpenI rows with non-empty findings text, the per-row weight distribution within the surviving 10,839 rows is 95.6% at $w_{\min} = 0.2$ and 4.4% at $w = 1.0$. The continuous-SFC ablation replaces the discrete $\{w_{\min}, 1\}$ rule with $\text{SFC}(x) = 1 - p_{\text{text}}^{\text{correct}}(x)$ clipped to $[w_{\min}, 1]$.

B.3. IMG_correct derivation and the “IMG_correct > IMG” identity

For a binary classifier f with gold class y , partition examples into cc/ci/ic/ii by whether f is correct on the full and image-zeroed inputs. Then $\text{IMG} = (n_{\text{ci}} - n_{\text{ic}})/n \times 100$. $\text{IMG}_{\text{correct}}$ further partitions ci into a margin-passing subset (where $p_{\text{full}}(y) - p_{\text{masked}}(y) \geq \tau$) and a margin-failing subset (the induced-inversion sub-population):

$$\begin{aligned} \text{IMG}_{\text{correct}} &= n_{\text{ci,with-margin}}/n \times 100, \\ \text{induced inv.} &= (n_{\text{ci,no-margin}} - n_{\text{ci,with-margin}})/n \times 100. \end{aligned}$$

$\text{IMG}_{\text{correct}}$ can exceed IMG. When the ic population (model becomes correct under masking) is non-trivial and the margin- failing ci population is small, $\text{IMG}_{\text{correct}} > (n_{\text{ci}} - n_{\text{ic}})/n$. This is why several configurations in the retroactive audit show $\text{IMG}_{\text{correct}} > \text{IMG}$; it is mathematically valid and does not indicate negative induced inversion. For autoregressive generators we apply this logic per token on the gold target sequence (with

Table 3. Training hyperparameters across all five configurations. The configurations differ only in (i) which weights JSONL is loaded, (ii) whether the faith loss is enabled, (iii) whether the flipped counterfactual is enabled, and (iv) the resulting λ_{faith} . All other hyperparameters are identical.

Hyperparameter	Value
Backbone	MedGemma-4B-IT (research-only licence)
LoRA rank / alpha	16 / 32
LoRA targets	q_proj, k_proj, v_proj, o_proj
Trainable parameters	11.9 M (0.28% of 4.31 B base)
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	0.0 on LoRA targets, 0.01 elsewhere
LR schedule	cosine to zero, 50-step linear warm-up
Batch size	2 (grad accum $\times 8$; effective 16)
Max optimizer steps	2,500
Mixed precision	bf16
Faith-loss margin (nats)	0.3
Image masking	per-channel mean of batch pixel values
λ_{faith}	0.3 when enabled, 0 otherwise
w_{min} (filter downweight)	0.2
Filter activation threshold	0.7 on $p_{\text{branch}}^{\text{correct}}$
Random seeds	{42, 1337, 9001}
GPU	NVIDIA H100 80 GB (Modal)
Wallclock per fine-tune	1.5–4.0 h (faith-loss configs $\sim 3\times$ slower)

the causal-LM shift between logits and target IDs).

B.4. Training hyperparameters and compute

Faith-loss configurations are $\sim 3\times$ slower per step than simple-loss configurations because each step runs three forward passes (full, image-zeroed, image-flipped). Total Phase 4 training compute: $\approx$ \$200 on Modal H100 80 GB across 15 fine-tunes (5 configs $\times$ 3 seeds). Total project envelope (training, primary evals, retroactive audit, cross-counterfactual, cross-site): $\approx$ \$400.

C. Full IMG_correct Retroactive Audit

Distribution of induced-inversion contributions. Across all 18 cells: induced inversion has an absolute range of [0.00, 1.31] pp and a relative range of [0.0%, 13.2%] of the corresponding IMG. The cell with the largest relative induced inversion is v5_0_base at 13.2%, with the smallest absolute IMG (1.28 pp); fractionally large because any flipping at all under masking is fractionally large at this scale. The cell with the largest absolute induced inversion is v6_0_loo_no_padchest at 1.31 pp (3.1% of IMG). No cell exceeds the pre-registered $\geq 30\%$ threshold.

The abl_no_consist cell is the cleanest disconfirmation of the pre-registered concern. Removing the consistency loss collapses IMG_correct from ~ 70 pp (nominal v5 family) to 3.60 pp — near baseline — while simultaneously reducing induced inversion from ~ 0.7 pp to 0.30 pp. The pre-registered concern was that consistency-loss training caused induced inversion. The empirical pattern is the opposite: with the consistency loss, IMG_correct is high and induced inversion is small; without it, both are small.

D. Cross-Counterfactual Detail

Baseline anchor. On the matching crop counterfactual, sft_only seed-42 yields IMG = 0.33 pp, IMG_correct = 2.80 pp at 73.82% accuracy, induced inversion = 0.26 pp ($n = 105,921$ tokens). The cross-counterfactual sft_full – sft_only IMG_correct gap on crop is $4.81 - 2.80 = 2.01$ pp; the gap on the trained-mean counterfactual is $74.00 - 2.38 = 71.62$ pp. The order-of-magnitude difference between trained-mean and crop is the operational meaning of the partial- circularity finding.

Table 4. Full IMG_correct retroactive audit, all 18 cells (nine verifier configurations + nine v5 ablations, each on $n = 2,974$ verifier rows). Acc full and Acc masked are accuracies under the full and image-zeroed inputs, respectively.

Config	IMG_corr	IMG	Inv	Inv/IMG	Acc full	Acc mask
Headline configurations.						
v5_0_base	3.83	1.28	0.17	13.2%	92.30	91.02
v5_1_ground	3.40	2.62	0.13	5.1%	91.22	88.60
v5_2_real	64.66	61.84	0.91	1.5%	92.60	30.77
v5_3_contrast	71.05	68.12	0.74	1.1%	93.01	24.88
v5_4_final	68.43	63.55	0.40	0.6%	91.69	28.14
v6_0_3site	74.51	70.21	0.07	0.1%	91.06	20.85
v6_0_loo_no_openi	38.20	34.20	0.00	0.0%	87.42	53.23
v6_0_loo_no_padchest	43.95	41.80	1.31	3.1%	92.30	50.50
v6_0_loo_no_chestx	18.59	14.53	0.20	1.4%	81.27	66.75
v5 ablations.						
abl_scale_25	69.07	65.64	0.71	1.1%	92.70	27.07
abl_scale_50	70.51	67.59	0.81	1.2%	92.80	25.22
abl_scale_100	66.24	62.81	0.61	1.0%	92.80	29.99
abl_hothresh_70	73.84	69.97	0.57	0.8%	92.60	22.63
abl_hothresh_80	77.98	74.78	0.74	1.0%	92.77	17.99
abl_no_contrast	63.28	60.02	0.87	1.4%	92.77	32.75
abl_no_ground	72.53	69.64	0.50	0.7%	92.97	23.34
abl_no_uncert	73.40	70.54	0.98	1.4%	93.17	22.63
abl_no_consist	3.60	2.79	0.30	10.8%	88.80	86.01

Table 5. Cross-counterfactual eval — full numbers including masked accuracy. Each row uses the same sft_full seed-42 checkpoint; only the counterfactual transformation differs.

Counterfactual	IMG	IMG_corr	Acc full	Acc mask
mean (TRAINED)	73.97	74.00	75.48	1.51
noise	62.10	62.20	75.48	13.38
crop	3.64	4.81	75.48	71.83
occlude	0.28	1.11	75.48	75.20

E. Cross-Site Detail

Silver target construction. ChestX-Det10 test rows in the standard test set have empty evidence_text fields. We constructed groundbench_test_chestx_det10_silver.jsonl (1,122 rows) by injecting CheXagent-2-3B (Chen et al., 2024) generations as reference_report; the silver-target nature is reported explicitly. PadChest-GR cross-site (Castro et al., 2024) requires a fresh CheXagent generation pass and is deferred. Token-level analysis. The cross-site 54.65% token-level accuracy is 20 pp lower than the in-distribution OpenI accuracy, reflecting (i) genuine cross-site difficulty (different camera vendors, different radiologist phrasing) and (ii) the silver-target nature (the model is being graded against another model’s output, not radiologist gold).

F. Pre-Registered Hypotheses

(i) Full method reduces IMG_correct on OpenI-test by ≥ 5 pp vs. SFT-only at matched accuracy. Status: HIT BY AN ORDER OF MAGNITUDE on three seeds $\times$ three faith-loss configurations. Realised: +72.58 pp at +1.17 pp accuracy.

(ii) IMG_correct refinement reveals $\geq 30\%$ of v5_3_contrast’s IMG as induced inversion. Status: REFUTED. Realised: 1.1%. Pattern holds across all 18 audited cells (induced-inversion contributions ≤ 1.31 pp absolute, $\leq 13.2\%$ relative).

(iii) Dual filter dominates the faith loss in the ablation. Status: INVERTED. The empirical pattern is the opposite: sft_dual is statistically indistinguishable from sft_only; the faith loss is the load-bearing component (sft_faith alone gives +71.73 pp over baseline). The increment from adding the dual filter on top of the faith loss is +0.85 pp seed-mean (within one standard deviation of sft_full). We retain the dual filter in the recipe because

the four-cell decomposition (App. B.2) demonstrates a symmetric-extension contribution axis not addressable by text-only filtering.

One pre-registration HIT, one REFUTED, one INVERTED — all reported transparently. The methodological refinement (IMG_correct) lands as a contribution independently of the empirical results.

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